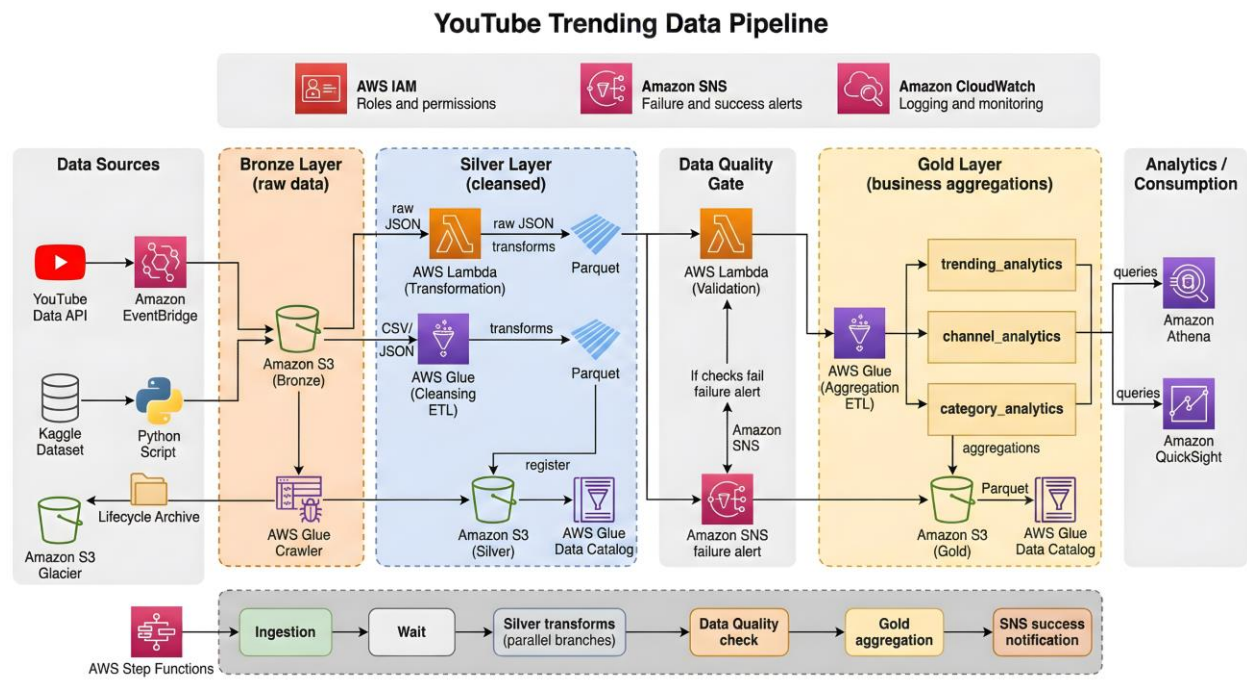


PRODUCTION READY DATA ENGINEERING

YOUTUBE TRENDING DATA PIPELINE

A cloud-native ETL pipeline that ingests YouTube trending video data across 10 regions, transforms it through a medallion architecture (Bronze > Silver > Gold), enforces data quality gates, and produces analytics-ready aggregations — all orchestrated by AWS Step Functions.



Overview:

This pipeline automates the end-to-end process of collecting, cleaning, and analyzing YouTube trending video data. It replaces manual Kaggle dataset downloads with live YouTube Data API v3 integration and produces three sets of business analytics tables:

- Trending Analytics — daily trending metrics per region (total videos, views, engagement rates)
- Channel Analytics — channel-level performance and ranking across regions
- Category Analytics — category-level breakdowns with view share percentages

The pipeline supports 10 regions and runs on a configurable schedule via AWS EventBridge.

Tech Stack: Component - Technology

AWS Global Infrastructure: Regions, AZ – Availability Zones (Data Centers or Servers), Edge Locations (to get content faster i.e low latency), AWS Accounts

Storage: S3, EBS, EFS

Databases: RDS, DynamoDB, Redshift & DMS

Compute: EC2, Lambda, AWS Glue, EMR

Analytics Services: Athena, Redshift Spectrum, Glue Catalog, QuickSight & Kinesis

AWS Global Infrastructure:

Edge Location:

Eg: If Netflix hosted their US based movies in US regions, if you want access them from India, then Edge Locations plays critical role instead accessing US servers.

IAM User:

A IAM user is the one who can access AWS console through the credentials created

Storage:

S3 buckets:

It is an object storage which has key value pairs.

Amazon EBS is an elastic block-level storage where volume attached to **EC2 instances** (kind of laptop in cloud).

EFS is an elastic file system. It is a fully managed NFS file system shared across multiple EC2 instances and AZ's.

Eg: Like how our laptops have harddisk (ROM), similarly, EC2 can have EBS and how we can attach external hard drives to our laptop for data transfer, like EFS for EC2.

Compute:

Amazon EC2

- Virtual servers in the cloud (Elastic Cloud Compute)
- Instance Types:
General Purpose, Compute Optimized, Memory Optimized, Storage Optimized
- Pricing: On-Demand, Reserved, Spot, Savings Plans
- AMI – pre-configured server templates
- Security Groups – virtual firewall for instances

AWS Lambda

- Serverless compute - run code without managing servers
- event driven: triggers from s3, API gateway, SQS

- Max 15 min execution, 10 GB memory
- Ideal for microservices, data processing and automation

AWS Glue:

- Fully managed serverless ETL service
- Glue Data Catalog –centralized metadata repository
- crawlers auto-discover schema from S3, JDBC etc..
- ETL jobs written in pyspark or python shell

AWS Glue Crawler:

An AWS Glue Crawler automatically scans data stores (like Amazon S3), infers schemas, detects data partitions, and populates the AWS Glue Data Catalog with metadata

Amazon EMR:

- Managed Big Data Platform (Hadoop, Spark, Hive, Presto)
- Process vast amount of data at scale
- Auto-scaling clusters with Spot Instance support
- EMR on EC2, EMR on EKS, or EMR Serverless
- Notebooks for interactive data exploration
- EMRFS enables direct S3 access at HDFS

EMRFS (Elastic MapReduce File System) is an implementation of the Apache Hadoop FileSystem interface that allows Amazon EMR (Elastic MapReduce) clusters to read and write data directly from Amazon S3.

Instead of storing data locally on the EMR cluster's EC2 instances (using HDFS), EMRFS turns Amazon S3 into your **persistent data lake layer**, separating computing power from data storage.

Query & Catalog Services (Analytics Service)

Amazon Athena:

- Serverless Interactive Query Service
- Standard SQL on S3 data
- Uses Glue Data Catalog as metastore
- CTAS for ETL and data transformation

Redshift Spectrum

- Datawarehouse OLAP service
- Query S3 data from Redshift Cluster
- Extends Redshift to exabyte scale

- Uses Redshift SQL + optimizer

Glue Data Catalog

- Centralized metadata store
- Tables, Schemas, partitions
- Shared across Athena, Redshift, EMR
- Crawlers auto-update schemas
- Hive compatible metastore
- Versioned table definitions

Note: AWS Glue Data Catalog stores only the metadata of the actual s3 data for clear schema info. If there

Amazon QuickSight:

- Cloud-native serverless BI Service
- Interactive dashboards and visualizations
- ML-powered insights (anomaly detection, forecasting)
- Data Sources: Athena, S3, Redshift, RDS and more
- Embedded analytics for apps.

Amazon Kinesis:

- Realtime streaming data collection and processing
 1. Kinesis Data Streams: Custom real-time processing with shards
 2. Kinesis Data Firehose: Managed delivery to S3, Redshift, OpenSearch
 3. Kinesis Data Analytics: SQL or Apache Flink on streaming data
- Usecases: IoT, clickstream, log analytics

Managed Database Services

Amazon RDS

- Managed relational database service
- Engines: MySQL, PostgreSQL, Oracle, SQL Server, MariaDB, Aurora
- Automated backups, patching, scaling
- Multi-AZ for high availability, Read Replicas for read scaling

Amazon DynamoDB:

- Fully managed NoSQL key-value and document database
- Single-digit millisecond latency at any scale
- On-demand or provisioned capacity modes
- Global Tables for multi-region replication

Amazon Redshift

- Petabyte-scale cloud data warehouse
- Columnar storage with massively parallel processing
- Redshift Spectrum queries S3 data directly
- RA3 nodes decouple compute from storage

AWS DMS

- Database Migration Service — migrate with minimal downtime
- Supports homogeneous and heterogeneous migrations
- Continuous data replication (CDC)
- Use with SCT for schema conversion

Other Services:

SNS: Amazon Simple Notification Service (Amazon SNS) is a fully managed pub/sub (publish/subscribe) messaging service designed for high-throughput, highly reliable message delivery. It allows applications to send notifications to thousands of subscribing endpoints or directly to end-users via SMS, email, or mobile push notifications.

CloudWatch: Cloudwatch is an intelligent observability and monitoring service designed to track application performance i.e metrics, logs, alarms, dashboards.

AWS Step Function:

AWS Step Function is a serverless orchestration service that lets you connect multiple AWS services and microservices into reliable, visual application workflows

Data Workflow:

Initially, create

S3 Buckets:

Create 3 buckets in S3 for storing data as per medallion architecture

Bronze Bucket Name - yt-data-pipeline-bronze-us-east-2-dev

Silver Bucket Name - yt-data-pipeline-silver-us-east-2-dev

Gold Bucket Name - yt-data-pipeline-gold-us-east-2-dev

Script Bucket - yt-data-pipeline-script-us-east-2-dev

IAM User:

Initially create a user for loading static Kaggle data into the raw layer i.e raw S3 bucket

The screenshot displays the AWS IAM console interface for a user named **aws_cli_local**. The left sidebar shows the navigation menu with sections for Identity and Access Management (IAM), Access Management, and Access reports. The main content area is titled **aws_cli_local** and includes a **Delete** button. The **Summary** section provides details about the user's ARN, console access status, creation date, and last console sign-in. It also lists two access keys, with the first one being active and used 9 days ago. The **Permissions** tab is selected, showing a table of permissions policies. The table lists one policy, **AdministratorAccess**, which is AWS managed and attached directly. A **Permissions boundary** section at the bottom indicates that no boundary is currently set.

Policy name	Type	Attached via
AdministratorAccess	AWS managed - job function	Directly

Under the Security Credentials tab, you can create an access key ID, secret access Key (credentials) for logging into AWS through AWS CLI (terminal).

Terminal:

```
mohammadshaik@macbookair ~ % aws configure
```

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

```
[AWS Access Key ID [None]: AKIAQ4WFIQGLAYBIEPL6
[AWS Secret Access Key [None]: qdceeAWXoq4V4gwiBYhfp//Tj9tLLvCNyAJAWpN7
[Default region name [None]: us-east-2
[Default output format [None]:
```

```
mohammadshaik@macbookair ~ % aws s3 ls
2026-08-21 18:35:34 rds-export-source-account-a
2026-04-28 15:13:04 rds-snapshot-export-demo-mohammad
2026-06-11 14:06:43 yt-data-pipeline-bronze-us-east-2-dev
2026-06-11 13:57:30 yt-data-pipeline-gold-us-east-2-dev
2026-06-11 13:58:47 yt-data-pipeline-script-us-east-2-dev
2026-06-11 13:57:10 yt-data-pipeline-silver-us-east-2-dev
mohammadshaik@macbookair ~ % cd /Users/mohammadshaik/Desktop/Mohammad/AWS/YouTube-Data-Pipeline
mohammadshaik@macbookair YouTube-Data-Pipeline % ls
data  scripts
mohammadshaik@macbookair YouTube-Data-Pipeline % aws s3 cp data/CAvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ca/
upload: data/CAvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ca/CAvideos.csv
mohammadshaik@macbookair YouTube-Data-Pipeline % aws s3 cp data/CA_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/
upload: data/CA_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/CA_category_id.json
mohammadshaik@macbookair YouTube-Data-Pipeline % clear
mohammadshaik@macbookair YouTube-Data-Pipeline % aws s3 cp data/CAvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ca/
aws s3 cp data/CA_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/
aws s3 cp data/DEvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=de/
aws s3 cp data/DE_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=de/
aws s3 cp data/FRvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=fr/
aws s3 cp data/FR_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=fr/
aws s3 cp data/GBvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=gb/
aws s3 cp data/GB_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=gb/
aws s3 cp data/INvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=in/
aws s3 cp data/IN_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=in/
aws s3 cp data/JPvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=jp/
aws s3 cp data/JP_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=jp/
aws s3 cp data/KRvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=kr/
aws s3 cp data/KR_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=kr/
aws s3 cp data/MXvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=mx/
aws s3 cp data/MX_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=mx/
aws s3 cp data/RUvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ru/
aws s3 cp data/RU_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ru/
aws s3 cp data/USvideos.csv s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=us/
aws s3 cp data/US_category_id.json s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=us/
upload: data/CAvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ca/CAvideos.csv
upload: data/CA_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/CA_category_id.json
upload: data/DEvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=de/DEvideos.csv
upload: data/DE_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=de/DE_category_id.json
upload: data/FRvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=fr/FRvideos.csv
upload: data/FR_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=fr/FR_category_id.json
upload: data/GBvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=gb/GBvideos.csv
upload: data/GB_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=gb/GB_category_id.json
upload: data/INvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=in/INvideos.csv
upload: data/IN_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=in/IN_category_id.json
upload: data/JPvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=jp/JPvideos.csv
upload: data/JP_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=jp/JP_category_id.json
upload: data/KRvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=kr/KRvideos.csv
upload: data/KR_category_id.json to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=kr/KR_category_id.json
upload: data/MXvideos.csv to s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=mx/MXvideos.csv
```

Src: kaggle data set csv and json

Target: s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ca/CAvideos.csv

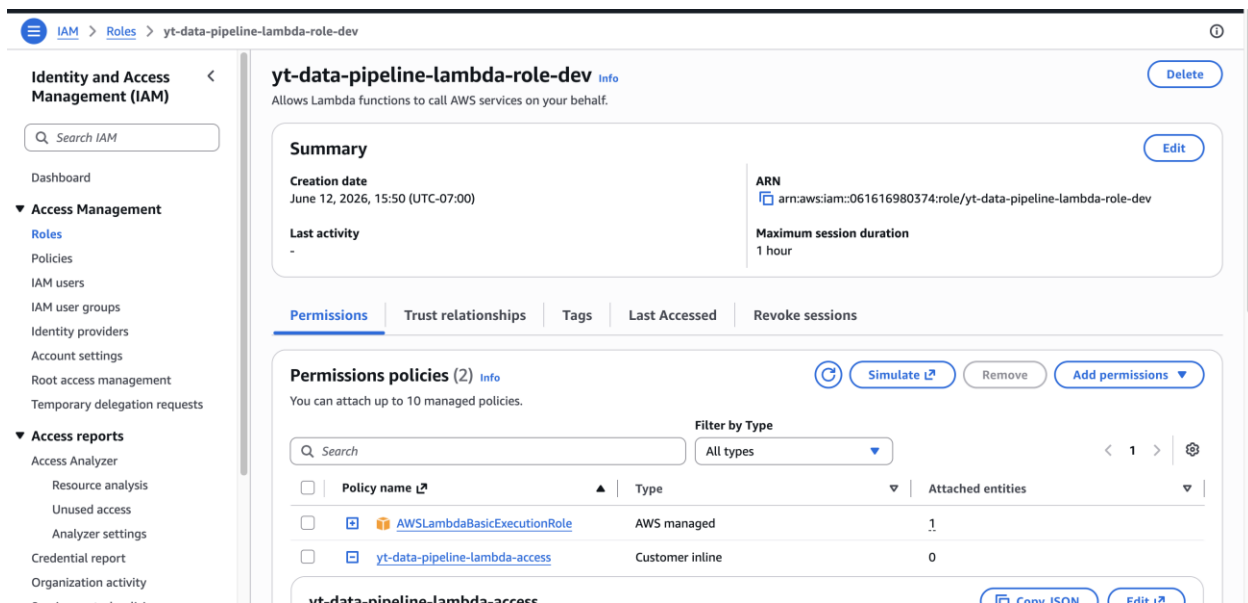
s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/CA_category_id.json

.....

IAM Role:

If the AWS services want to communicate with each other, IAM Role is the one that can have permissions attached to each other's services.

yt-data-pipeline-lambda-role-dev:



yt-data-pipeline-lambda-access:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "S3Access",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::yt-data-pipeline-bronze-us-east-2-dev/*",
        "arn:aws:s3:::yt-data-pipeline-bronze-us-east-2-dev",
        "arn:aws:s3:::yt-data-pipeline-silver-us-east-2-dev/*",
        "arn:aws:s3:::yt-data-pipeline-silver-us-east-2-dev"
      ]
    },
    {
      "Sid": "GlueAccess",
      "Effect": "Allow",
      "Action": [
        "glue:GetTable",
```



```
        "glue:GetDatabase",
        "glue:CreateTable",
        "glue:UpdateTable",
        "glue:GetPartitions",
        "glue:CreatePartition",
        "glue:BatchCreatePartition"
    ],
    "Resource": "*"
},
{
    "Sid": "SNSAccess",
    "Effect": "Allow",
    "Action": "sns:Publish",
    "Resource": "arn:aws:sns:us-east-2:061616980374:yt-data-pipeline-alerts-dev:5560a8c0-33a1-4807-96be-b486522ca0da"
},
{
    "Sid": "AthenaAccess",
    "Effect": "Allow",
    "Action": [
        "athena:StartQueryExecution",
        "athena:GetQueryExecution",
        "athena:GetQueryResults"
    ],
    "Resource": "*"
}
]
```

yt-data-pipeline-glue-role-dev

Navigation: IAM > Roles > yt-data-pipeline-glue-role-dev

yt-data-pipeline-glue-role-dev info

Allows Glue to call AWS services on your behalf.

[Delete](#)

Summary

Creation date
June 23, 2026, 01:05 (UTC-05:00)

Last activity
4 weeks ago

ARN
arn:aws:iam::061616980374:role/yt-data-pipeline-glue-role-dev

Maximum session duration
1 hour

[Edit](#)

[Permissions](#) | [Trust relationships](#) | [Tags](#) | [Last Accessed](#) | [Revoke sessions](#)

Permissions policies (2) info

You can attach up to 20 managed policies.

[Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type: All types

☐	Policy name info	Type	Attached entities
☐	AWSGlueServiceRole	AWS managed	1
☐	yt-data-pipeline-glue-access	Customer inline	0

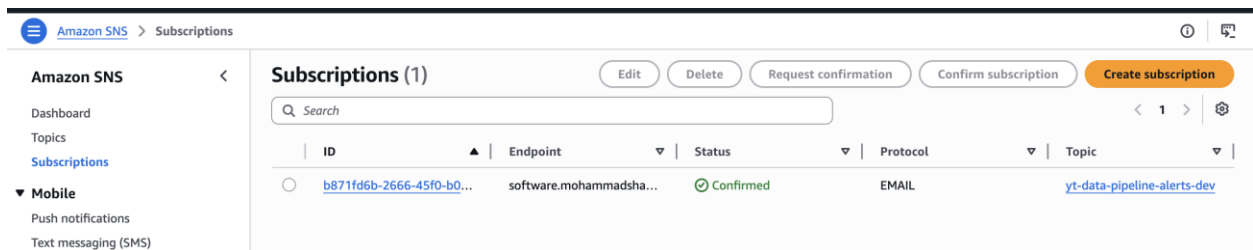
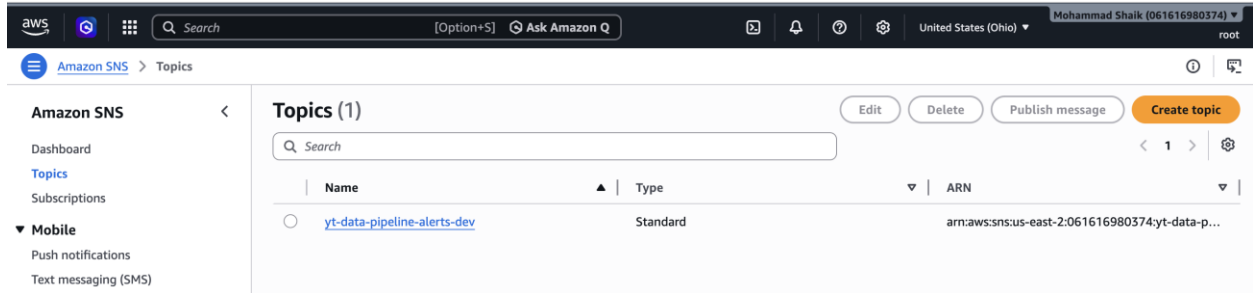
[Copy JSON](#) [Edit](#)

yt-data-pipeline-glue-access:

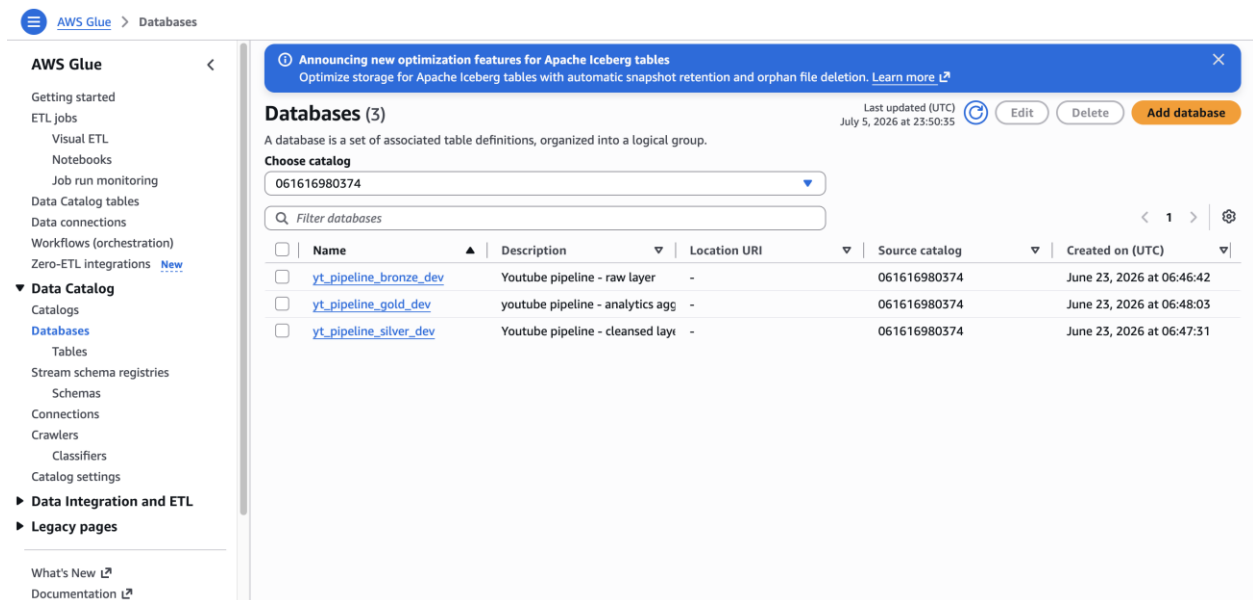
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "S3FullAccess",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:ListBucket",
        "s3:DeleteObject"
      ],
      "Resource": [
        "arn:aws:s3::yt-data-pipeline-bronze-us-east-2-dev/*",
        "arn:aws:s3::yt-data-pipeline-bronze-us-east-2-dev",
        "arn:aws:s3::yt-data-pipeline-silver-us-east-2-dev/*",
        "arn:aws:s3::yt-data-pipeline-silver-us-east-2-dev",
        "arn:aws:s3::yt-data-pipeline-gold-us-east-2-dev/*",
        "arn:aws:s3::yt-data-pipeline-gold-us-east-2-dev",
        "arn:aws:s3::yt-data-pipeline-scripts-us-east-2-dev/*",
        "arn:aws:s3::yt-data-pipeline-scripts-us-east-2-dev"
      ]
    }
  ]
}
```

SNS service:

Created a topic (publish) which Sends notifications to the subscribed email or text.



Glue Data Catalog: Created 3 databases



AWS Glue Crawler:

yt-data-pipeline-bronze-crawler-dev: Scans the bronze s3 data and restores into Data Catalog

Src:

s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/

s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/

Target:

yt_pipeline_bronze_dev.raw_statistics,

yt_pipeline_bronze_dev.raw_statistics_reference_data

AWS Glue

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Zero-ETL integrations

New

▼ Data Catalog

Catalogs

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

▼ Data Integration and ETL

Zero-ETL integrations

New

ETL jobs

Visual ETL

Notebooks

yt-data-pipeline-bronze-crawler-dev

Last updated (UTC)
July 25, 2026 at 01:22:54

Run crawler

Edit

Delete

Crawler properties

Name

yt-data-pipeline-bronze-crawler-dev

IAM role

yt-data-pipeline-glue-role-dev

↗

Database

yt_pipeline_bronze_dev

State

READY

Description

-

Security configuration

-

Lake Formation configuration

-

Table prefix

-

Maximum table threshold

-

▶ Advanced settings

Crawler runs (1)

The list of crawler runs for this crawler.

Filter data

Filter by a date and time range

Start time (UTC)

▲

End time (UTC)

▼

Current/last duration

▼

Status

▼

DPU hours

▼

Table changes

▼

June 23, 2026 at 06:55:54

June 23, 2026 at 06:56:45

50 s

Completed

0.150

2 table changes, 20 partition changes

AWS Glue

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Zero-ETL integrations

New

▼ Data Catalog

Catalogs

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Stream schema registries

Schemas

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Classifiers

Catalog settings

▼ Data Integration and ETL

Zero-ETL integrations

New

ETL jobs

Visual ETL

Notebooks

Announcing new optimization features for Apache Iceberg tables

Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion.

Learn more

Tables

A table is the metadata definition that represents your data, including its schema. A table can be used as a source or target in a job definition.

Tables (2)

Last updated (UTC)
July 25, 2026 at 01:23:54

Delete

Add tables using crawler

Add table

View and manage all available tables.

Choose catalog

061616980374

Choose database

yt_pipeline_bronze_dev

Filter tables

Name

▲

Database

▼

Location

▼

Classifica...

▼

Deprecated

▼

View data

Data quality

Column st...

raw_statistics

yt_pipeline_bronz

s3://yt-data-pipel

CSV

-

Table data

View data quality

View statistics

raw_statistics_refe

yt_pipeline_bronz

s3://yt-data-pipel

JSON

-

Table data

View data quality

View statistics

Amazon Athena

Query editor tabs

Query 1

Query 2

Query 3

Query in Amazon SageMaker Unified Studio

Use your IAM role to analyse and build with your existing AWS resources in a single data and AI development environment.

Set up

Editor

Recent queries

Saved queries

Query settings

Workgroup

primary

Athena now supports typeahead code suggestions to speed up SQL query development

Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.

Edit preferences

Data

Data source

AwsDataCatalog

Catalogue

None

Database

yt_pipeline_bronze_dev

Tables and views

Create

Filter tables and views

Tables (2)

raw_statistics

Partitioned

raw_statistics_reference_data

Partitioned

Views (0)

Query 2

Query 3

1

SELECT * FROM "yt_pipeline_bronze_dev"."raw_statistics_reference_data" limit 10;

SQL

Ln 1, Col 1

Run again

Explain

Cancel

Clear

Create

Reuse query results

up to 60 minutes ago

Query results

Query status

Failed

Time in queue: 107 ms

Run time: 675 ms

Data scanned: -

HIVE_CURSOR_ERROR: Failed to read file at s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region-ru/RU_category_id.json

This query ran against the "yt_pipeline_bronze_dev" database, unless qualified by the query. Please post the error message on our [forum](#) or contact [customer support](#) with Query ID: 76468441-9c03-4609-818a-a357ae7f75db

Lambda:

The lambda function Ingest data from youtube api into the raw bucket.

Initially, updated the general config i.e timeout, memory etc. And add env variables. Now added the actual ingestion code in Code tab, then deployed and tested by adding a test event named “yt_bronze_test”

The lambda function ingestion replaces the “download from Kaggle and aws s3 cp” workflow with a real, automated, live data ingestion pipeline

Src: youtube API for 4 regions US,GB,CA,IN

Target:

s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics/region=ca/date=2026-07-06/hour=00/20260706_002352.json

s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/date=2026-07-06/ca_category_id.json

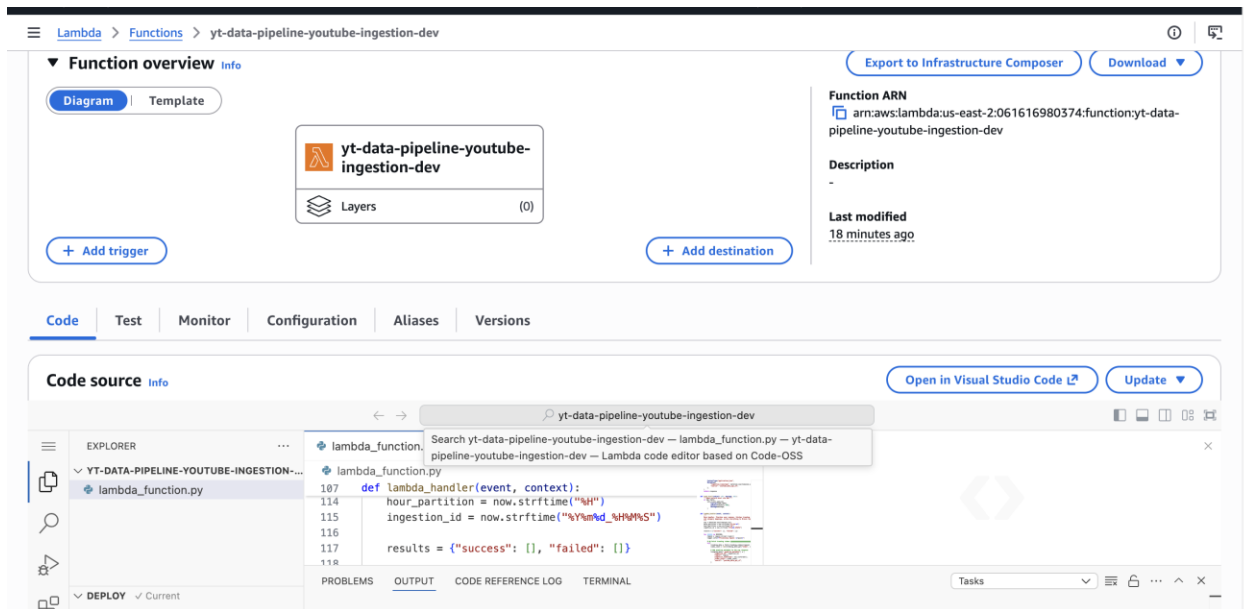
.....

The target both raw_statistics, raw_statistics_reference_data (videos and categories) will be json files.

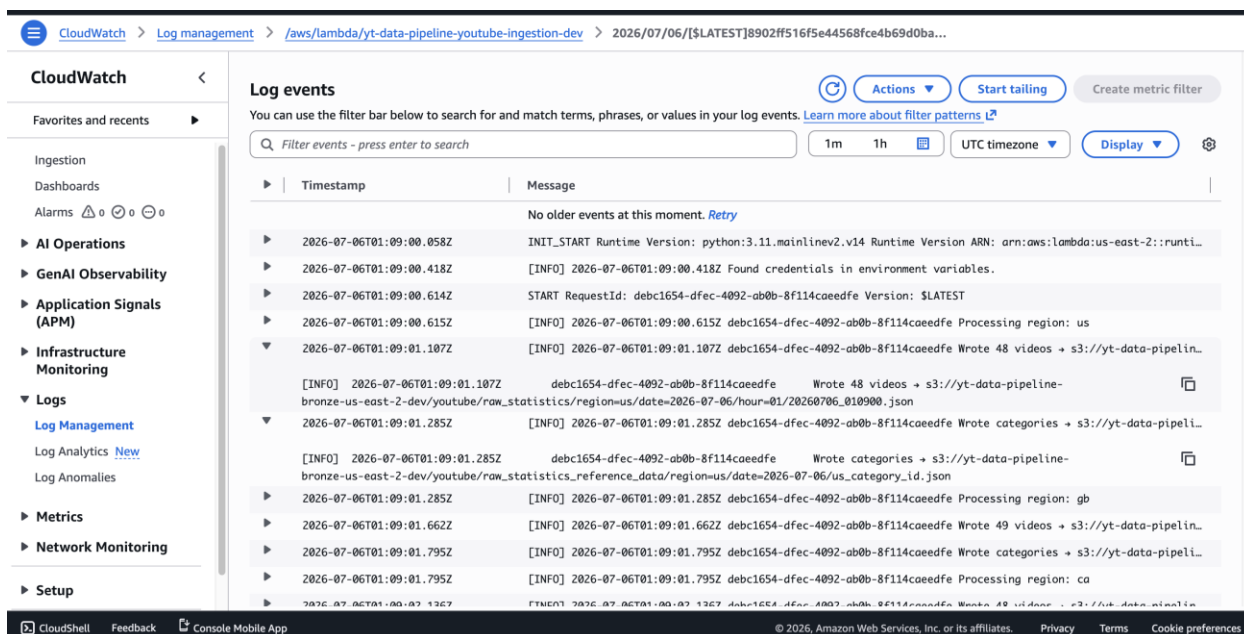
The screenshot displays the AWS Lambda console interface for the function `yt-data-pipeline-youtube-ingestion-dev`. The top section shows the function's basic details: Function ARN (`arn:aws:lambda:us-east-2:061616980374:function:yt-data-pipeline-youtube-ingestion-dev`), Description, and Last modified time (20 minutes ago). Below this, the **Configuration** tab is selected, showing the **Environment variables (4)** section. A search bar is present above a table of environment variables. The table lists the following variables:

Key	Value
<code>S3_BUCKET_BRONZE</code>	<code>yt-data-pipeline-bronze-us-east-2-dev</code>
<code>SNS_ALERT_TOPIC_ARN</code>	<code>arn:aws:sns:us-east-2:061616980374:yt-data-pipeline-alerts-dev:b871fd6b-2666-45f0-b076-13979650f702</code>
<code>YOUTUBE_API_KEY</code>	<code>AlzaSyAyXYho5JyGXIXC-4Hs8XEL5Z5ksSc9XHM</code>
<code>YOUTUBE_REGIONS</code>	<code>US,GB,CA,IN</code>

The left sidebar shows the navigation menu with options like General configuration, Triggers, Permissions, Destinations, Function URL, Environment variables (selected), and Tags.



Cloudwatch: we can find Output log here where it shows that the code ingested youtube API data (videos and categories) into bronze bucket



yt-data-pipeline-json-to-parquet-dev

The lambda functions trigger the event which has category data in bronze bucket that got ingested from the above lambda pipeline and does the transformation and finally loads the data into a silver bucket.

Initially, create a lambda pipeline and update/add the general config, env variables, code, deploy and test. Also, add the trigger to the function and then pandas to Layers as code contains pandas.

Here the functions overwrite the data in O/P target

Src: 4 regions US,GB,CA,IN only reference_data categories

s3://yt-data-pipeline-bronze-us-east-2-dev/youtube/raw_statistics_reference_data/region=ca/date=2026-07-06/ca_category_id.json

.....

Target: writes to silver layer i.e Glue Data Catalog table, S3 bucket

yt_pipeline_silver_dev.clean_reference_data

s3://yt-data-pipeline-silver-us-east-2-dev/youtube/reference_data/region=ca/4e3c9cc487554314a86321a16519ac01.snappy.parquet

The screenshot displays the AWS Lambda console interface for the function `yt-data-pipeline-json-to-parquet-dev`. The **Function overview** tab is active, showing a diagram of the function's configuration. The function is linked to an **S3** bucket as a trigger. The **Function ARN** is `arn:aws:lambda:us-east-2:061616980374:function:yt-data-pipeline-json-to-parquet-dev`. The **Description** field is empty. The **Last modified** timestamp is `30 minutes ago`. The **Code source** tab is also visible, showing the `lambda_function.py` file. The code includes environment variables for `SNS_ALERT_TOPIC_ARN` and `SNS_TOPIC_FOR_ALERTS`. The console footer shows the copyright notice: `© 2026, Amazon Web Services, Inc. or its affiliates.`

Lambda > Functions > yt-data-pipeline-json-to-parquet-dev

Layers (1)

S3

+ Add trigger

+ Add destination

Last modified 3 weeks ago

Code | Test | Monitor | Configuration | Aliases | Versions

Configuration

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

VPC

RDS databases

Triggers (1)

Search for triggers

Trigger

S3: yt-data-pipeline-bronze-us-east-2-dev

arn:aws:s3::yt-data-pipeline-bronze-us-east-2-dev

Details

Bucket arn: arn:aws:s3::yt-data-pipeline-bronze-us-east-2-dev

Event types: s3:ObjectCreated:*

isComplexStatement: No

Notification name: 674f152c-dcd4-4384-a103-633d328c7b54

Prefix: youtube/raw_statistics_reference_data/

Service principal: s3.amazonaws.com

Source account: 061616980374

Statement ID: lambda-a27b646f-1be4-4ee7-b39a-51909c656466

Suffix: json

CloudWatch > Log management > /aws/lambda/yt-data-pipeline-json-to-parquet-dev > 2026/07/06/[\$LATEST]1e4b6c7bd81547b3b093b8663202...

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Analytics

Log Anomalies

Metrics

Network Monitoring

Setup

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events.

Filter events - press enter to search

1m 1h

UTC timezone

Display

Log events table with columns: Timestamp, Message. Includes log entries for INIT_START, INFO Found credentials, START RequestId, END RequestId, and REPORT RequestId.

CloudShell | Feedback | Console Mobile App | © 2026, Amazon Web Services, Inc. or its affiliates. | Privacy | Terms | Cookie preferences

s3://yt-data-pipeline-silver-us-east-2-dev/youtube/statistics/region=ca/

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yt-data-pipeline-bronze-to-silver-dev

Last modified on 7/5/2026, 7:28:02 PM

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Name

yt-data-pipeline-bronze-to-silver-dev

Description - optional

Descriptions can be up to 2048 characters long.

IAM role

To access other AWS services and resources your Glue ETL job needs an IAM role. Choose a role with the right permissions for your Glue ETL job.

yt-data-pipeline-glue-role-... Create new role

Edit role

View role details in IAM

Type

The type of ETL job. This is set automatically based on the types of data sources you have selected.

Spark

Glue version

Q help

Glue 5.1 - Supports spark 3.5, Scala 2, Python 3

Language

Python 3

Worker type

Set the type of predefined worker that is allowed when a job runs.

G 1X (4vCPU and 16GB RAM)

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Job parameters

Q help

Key

Value - optional

--bronze_database X

yt_pipeline_bronze_dev X Remove

--bronze_table X

raw_statistics X Remove

--conf X

spark.eventLog.rolling.enabled= X Remove

--silver_bucket X

yt-data-pipeline-silver-us-east- X Remove

--silver_database X

yt_pipeline_silver_dev X Remove

--silver_table X

clean_statistics X Remove

Add new parameter

You can add 44 more parameters.

Amazon QuickSight

AWS Glue

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► **Data Integration and ETL**
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What's New ⓘ
Documentation ⓘ

yt-data-pipeline-bronze-to-silver-dev Last modified on 7/5/2026, 7:28:02 PM **Actions** **Save** **Run**

Script | Job details | **Runs** | Data quality | Schedules | Version Control

Job runs (1/1) Info
Last updated (UTC) July 6, 2026 at 02:39:42 **View details** Stop job run Troubleshoot with AI **Table View** Card View

Filter job runs by property

Run status	Retries	Start time (Local)	End time (Local)	Duration	Capacity ...	Worker type
Succeeded	0	07/05/2026 19:28:13	07/05/2026 19:30:55	2 m 34 s	10 DPUs	G.1X

Run details | Input arguments (16) | Logs | Run insights | Metrics | Troubleshooting analysis | Spark UI

Job name	Start time (Local)	DPU Hours	Glue version
yt-data-pipeline-bronze-to-silver-dev	07/05/2026 19:28:13	0.429444	5.1
Last modified on (Local)	Id	End time (Local)	Worker type
07/05/2026 19:30:55	jr_56b45d9f9387dd465d3a4c16ecd445d02760bb15b566bf00beea333aa10544e9	07/05/2026 19:30:55	G.1X
Log group name	Run status	Start-up time	Max capacity
/aws-glue/jobs	Succeeded	7 seconds	10 DPUs
Number of workers	Retry attempt number	Execution time	Execution class

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Lambda Data quality Job: **yt-data-pipeline-data-quality-dev**

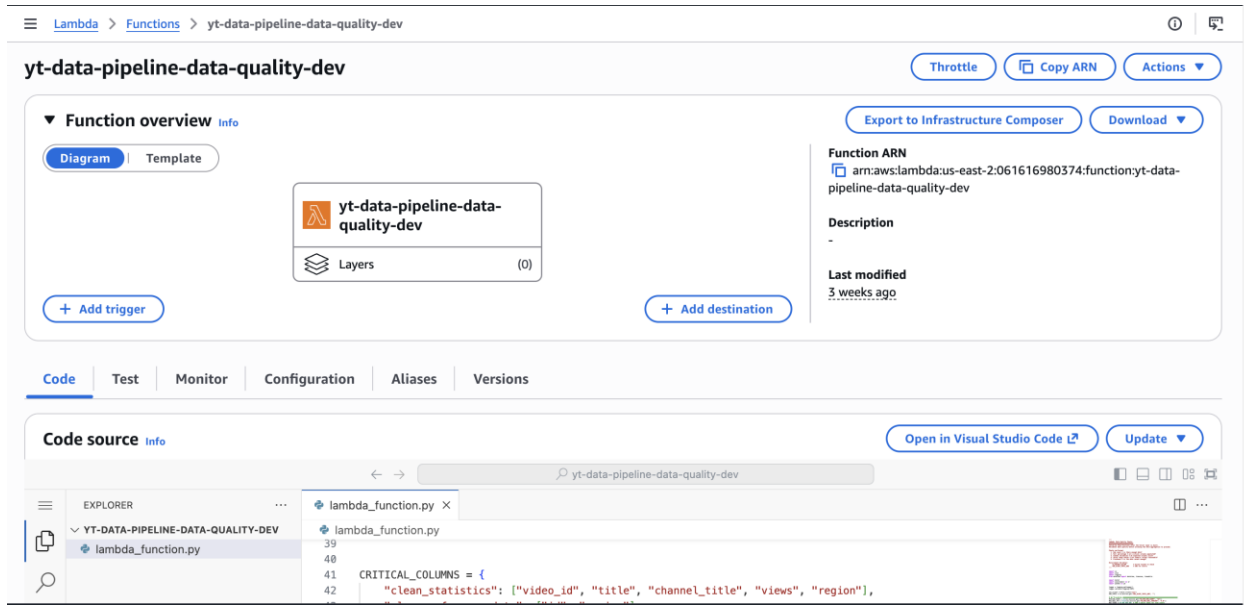
This Lambda function (3rd job) is related to Data Quality Checks and is Called by Step Functions after the Silver layer is built. It validates data quality before allowing the Gold aggregation to proceed.

Src: Silver layer Glue Catalog tables -

yt_pipeline_silver_dev. clean_statistics

yt_pipeline_silver_dev.clean_reference_data

Note: The code inside the lambda will execute a query using Athena to validate data quality checks which requires the lambda Iam role to grant access to Athena along with required permissions i.e access to Athena bucket for storing results etc..



Glue Job: **yt-data-pipeline-silver-to-gold-dev**

Silver → Gold (Analytics Aggregations)

The job Reads cleansed statistics and reference data from Silver, joins them and produces business-level aggregations in the Gold layer.

Gold layer tables are optimized for analytics queries in Athena/QuickSight.

Gold tables produced:

1. **Trending Analytics** — daily trending metrics per region (total videos, views, engagement rates)
2. **Channel Analytics** — channel-level performance and ranking across regions
3. **Category Analytics** — category-level breakdowns with view share percentages

Src: Silver layer Glue Catalog tables -

yt_pipeline_silver_dev.clean_statistics

yt_pipeline_silver_dev.clean_reference_data

Target: Gold layer Glue Catalog Tables and S3

yt_pipeline_gold_dev.trending_analytics — Daily trending summaries per region

yt_pipeline_gold_dev.channel_analytics — Channel performance metrics

yt_pipeline_gold_dev.category_analytics — Category-level trends over time

Step Functions:

Created a Step function which can orchestrate the pipeline end-to-end

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yt-data-pipeline-orchestration

Details

Arn

arn:aws:states:us-east-2:061616980374:stateMachine:yt-data-pipeline-orchestration

IAM role ARN

arn:aws:iam::061616980374:role/service-role/StepFunctions-yt-data-pipeline-orchestration-role-7myncekqu

Type

Standard

Status

Active

Creation date

Jul 31, 2026, 20:21:36 (UTC-07:00)

X-Ray tracing

Disabled

Executions

Monitoring

Logging

Definition

Aliases

Versions

Tags

Executions (0/5)

View details

Stop execution

Redrive

Start execution

Filter executions by property or value

All

Last 15 months

Local timezone

5 matches

< 1 >

☐	Name	Status	Start Time (local)	End Time (local)	Duration	Info
☐	0e00540d-3348-45b8-ba2f-8a3238aaaa17	Succeeded	Jul 31, 2026, 22:33:08	Jul 31, 2026, 22:40:28	00:07:19.855	-
☐	14330c58-ea3b-450f-88cf-fc01c3df5098	Succeeded	Jul 31, 2026, 21:59:52	Jul 31, 2026, 22:04:10	00:04:18.238	-

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Info

MOST POPULAR

AWS Lambda

Invoke

Amazon SNS

Publish

Amazon ECS

RunTask

AWS Step Functions

StartExecution

AWS Glue

StartJobRun

Bedrock AgentCore

InvokeHarness



Workflow

Definition

The top level state machine properties for this workflow. Learn more

State machine query language

Info

JSONPath

Start at

The state that is the starting point of the workflow.

IngestFromYouTubeAPI

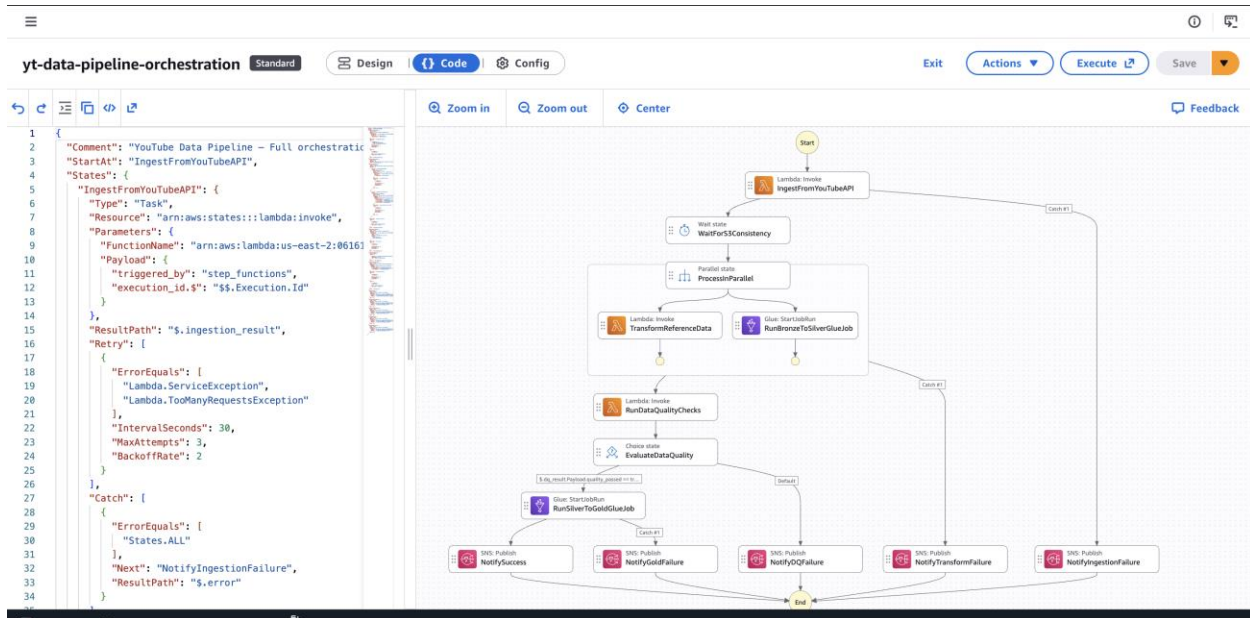
Comment - optional

A human-readable description of the state machine.

YouTube Data Pipeline — Full orchestration via Step Functions. Replaces manual execution of individual jobs.

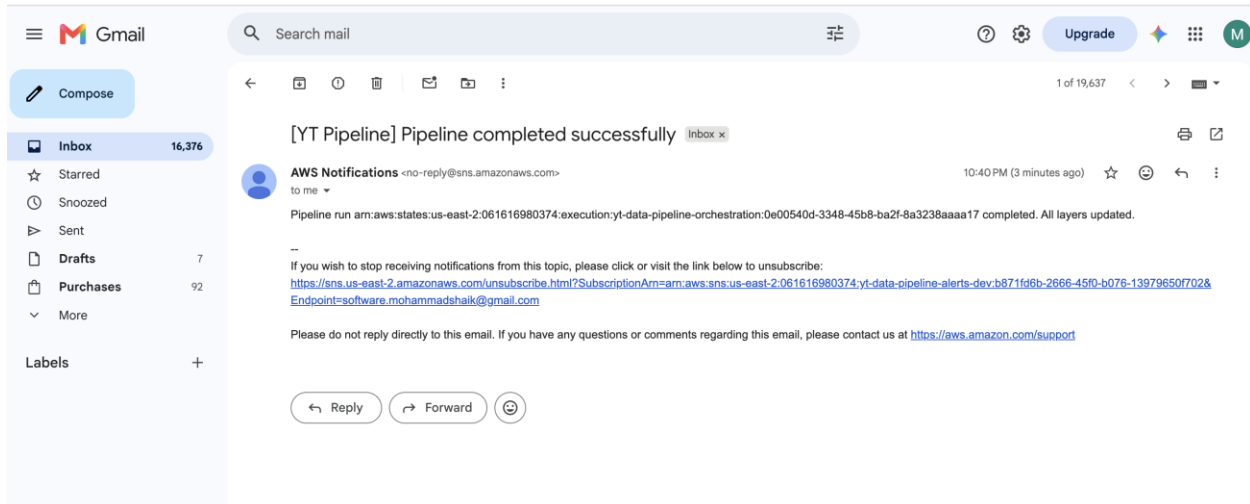
TimeoutSeconds - optional

The maximum number of seconds an execution of the state machine can run. If it runs longer than the specified time, the execution fails with a States.Timeout.

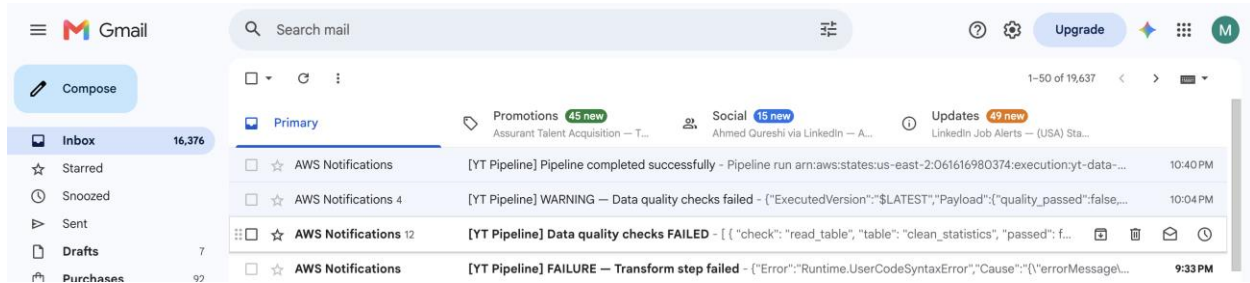


You can start the execution and monitor the pipeline and troubleshoot/debug the issues accordingly if it throws/raises any failure alerts.

SNS Alert Success:



Also received many Failures Alerts:



But resolved accordingly using AI.

Ref: <https://github.com/darshilparmar/youtube-data-pipeline-aws-s3-lambda-glue-athena-stepfunction/blob/main/README.md>